SeeWandel Climate

Interreg
Alpenrhein-Bodensee-Hochrhein

Kofinanziert von der Europäischen Union

# Regime Shifts in Plankton Dynamics: What Drives Them and Can We Predict Them?

Dr. Leonardo Capitani
Department of Aquatic Ecology, Eawag · SeeWandel-Climate

www.seewandel.org

Ecomeets @ Eawag, Dübendorf
28 October 2026

# Open science: take the slides with you

Scan to download the slides

SORTEE
Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology

Slides, code and data shared openly
leomarameo7.github.io/science_communication.html

# Acknowledgements




Pomati lab
Plankton Ecology, Eawag



Steve Munch & Tanya Rogers
EDM workshop guests

SeeWandel Climate

SeeWandel-Climate

Interreg
Alpenrhein-Bodensee-Hochrhein

Kofinanziert von der Europäischen Union

Supported by the
Biodiversity Center WSL–Eawag

eawag aquatic research

WSL

# Take home message

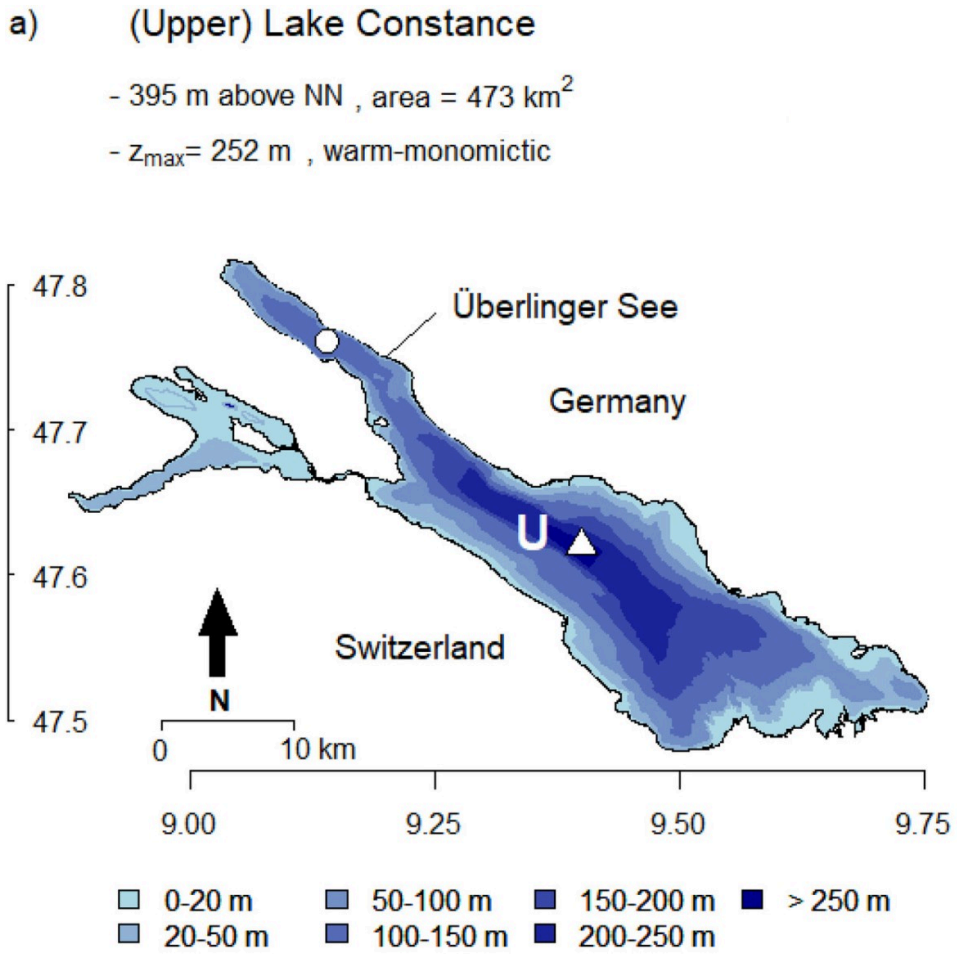
Plankton regime shifts can be detected and partly explained by biotic drivers. Whether they propagate and can be predicted is still open.

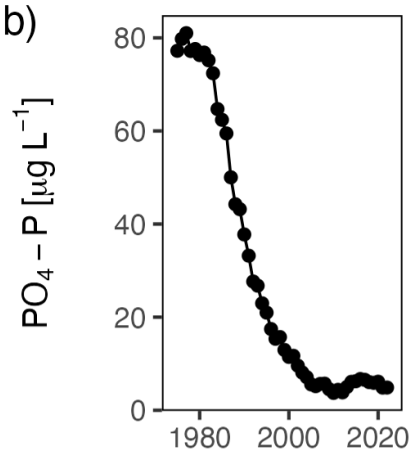


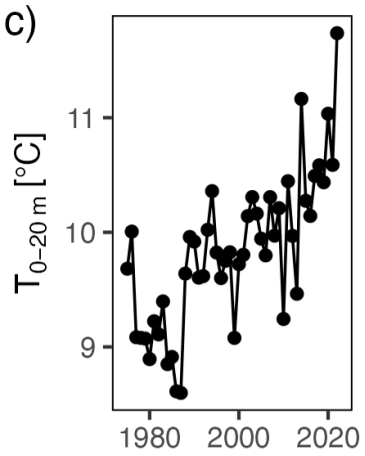
# Outline

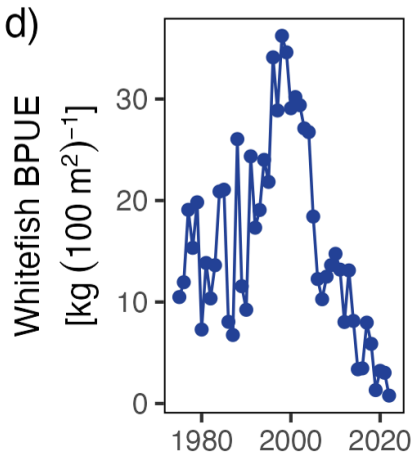
1. What we know: background
2. Research questions
3. Methods: our pipeline
4. Results: detection of regime shifts
5. Explain regime shifts
6. Propagation of regime shifts
7. Conclusions

# What we know: background



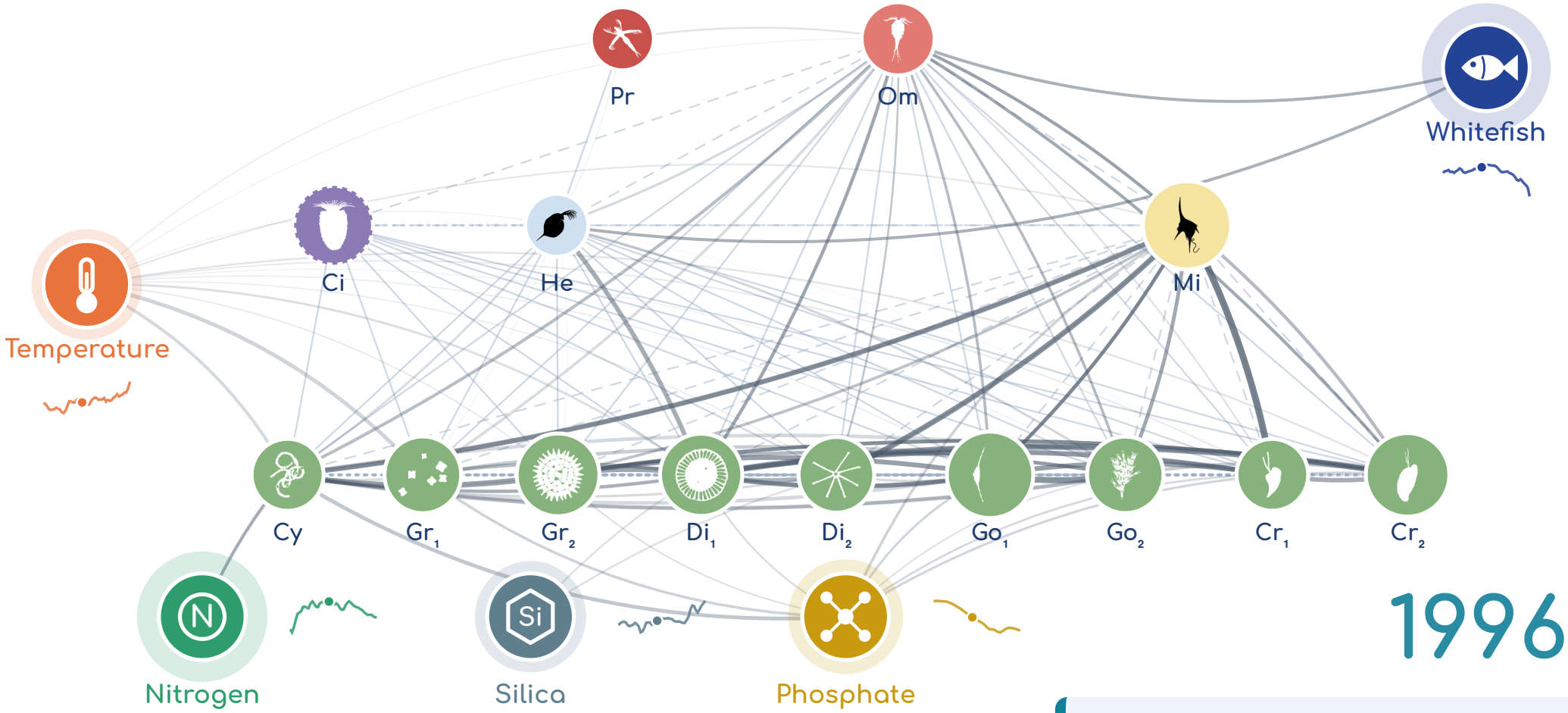




Lake Constance
473 km² · 252 m deep

Phosphate ↑ until 1980, then ↓

Temperature ↑ about 1 °C

# What we know: the plankton community




1996

Study period: 1975–2022 · monthly time series

The plankton community changes across time

Network modified from Merz et al. 2023, Nat. Clim. Change.

# Research questions



## 1 · Detect

Abrupt shifts?



## 2 · Explain

What drives them?



## 3 · Propagate

Does a shift spread?



## 4 · Predict

Can we forecast them? (in progress)

# Methods: our pipeline

1

## Data

Monthly data 1975–2022

2

## Detect

BEAST + two criteria

3

## Explain

GP-EDM + hypotheses

4

## Propagate

Shift-removal test

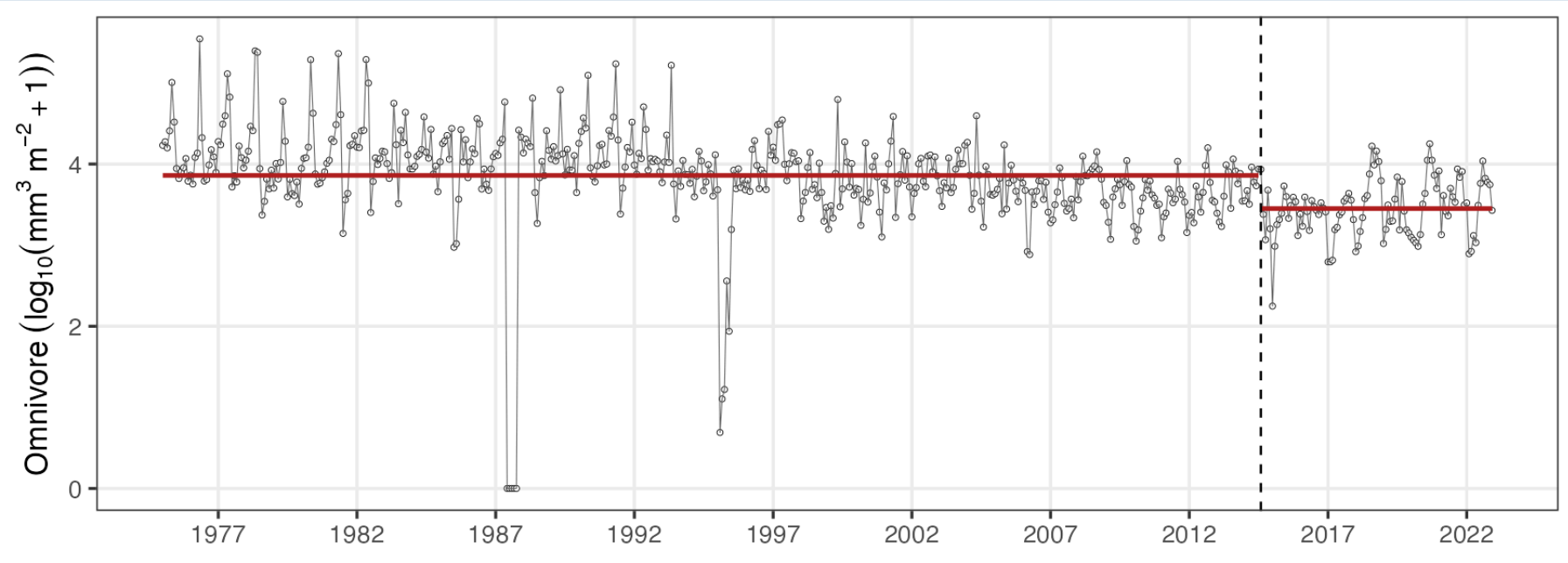
5

## Predict

Forward validation

WORK IN PROGRESS

# Results: detection of regime shifts




Omnivores, monthly biovolume. Dashed: shift; red: period means.

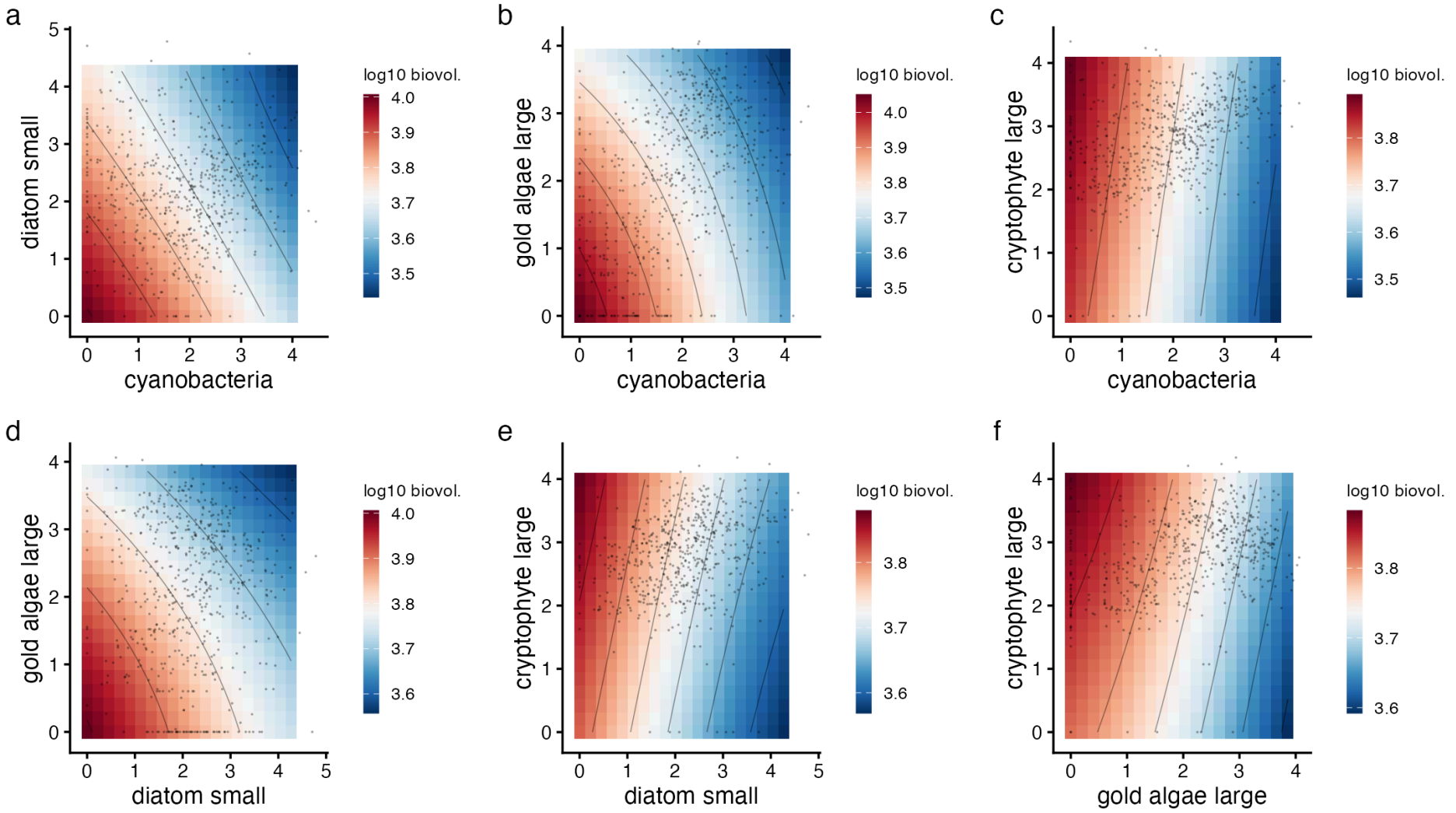


Shift: Aug 2014

abruptness 1.19

The omnivores dropped abruptly in 2014

# Explain regime shifts



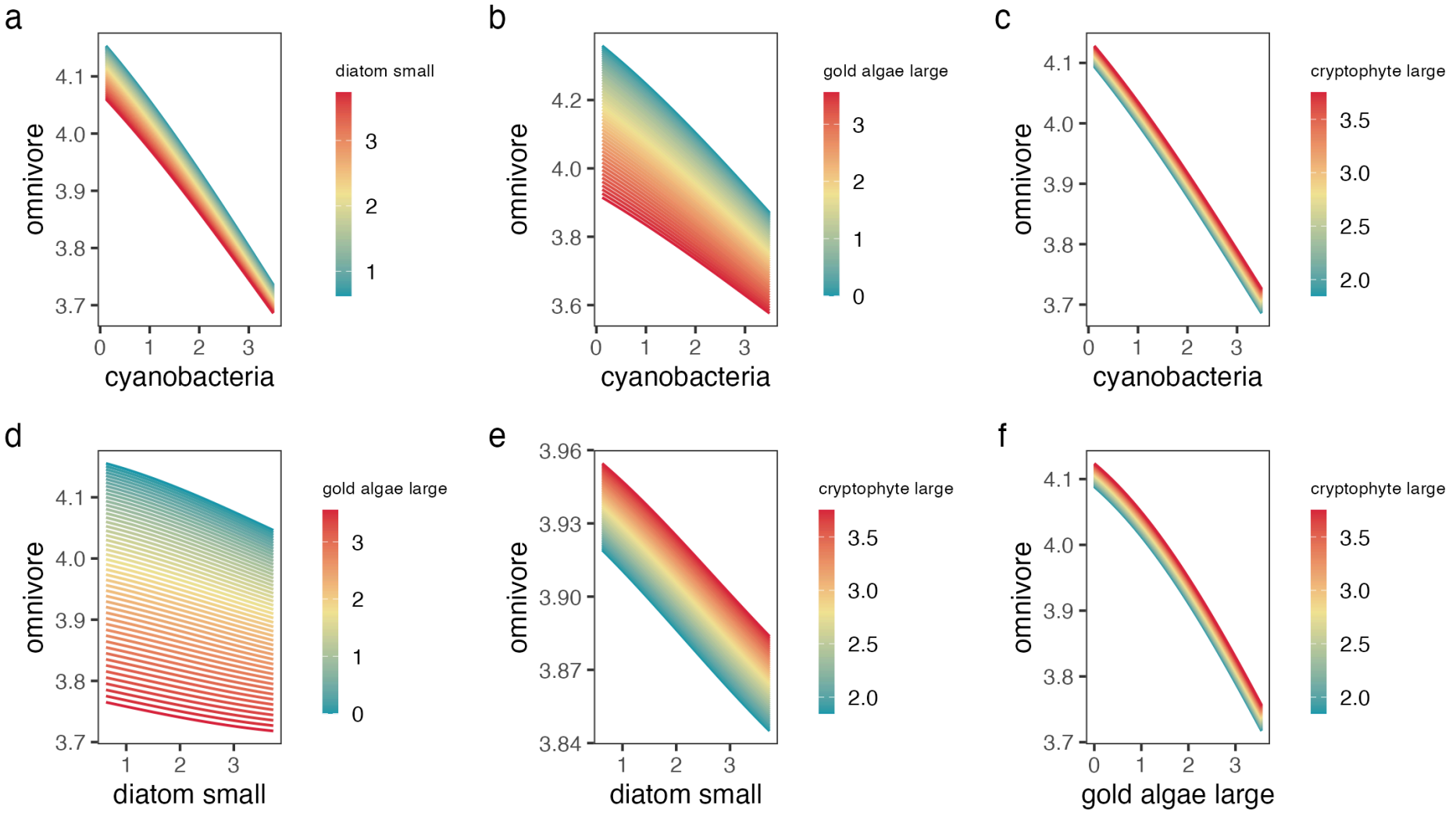



LOO $R^2$ = **0.81**

Omnivores ↓ with cyanobacteria, small diatoms, large gold algae

Prey hypothesis: **not supported**

# Explain regime shifts: partial dependence






x: predictor 1
lines: predictor 2

Cyanobacteria ↓ omnivores

Lines never cross

# Propagation of regime shifts

### 1 · Real predictor




2 candidate pairs

0 testable

Predictor shifted **after**

### 2 · Shift-free surrogate (× 100)




Gap between real and surrogate prediction = evidence for propagation

**No evidence yet: too few shifts**

# Conclusions



Shifts are detectable



Biotic drivers matter



Propagation, prediction: open

# Take home message

Plankton regime shifts can be detected and partly explained by biotic drivers. Whether they propagate and can be predicted is still open.



# Acknowledgements

Interreg
Alpenrhein-Bodensee-Hochrhein

Kofinanziert von der Europäischen Union

igkb
Internationale Gewässerschutzkommission für den Bodensee

IBKF

Supported by the Biodiversity Center WSL–Eawag

eawag aquatic research

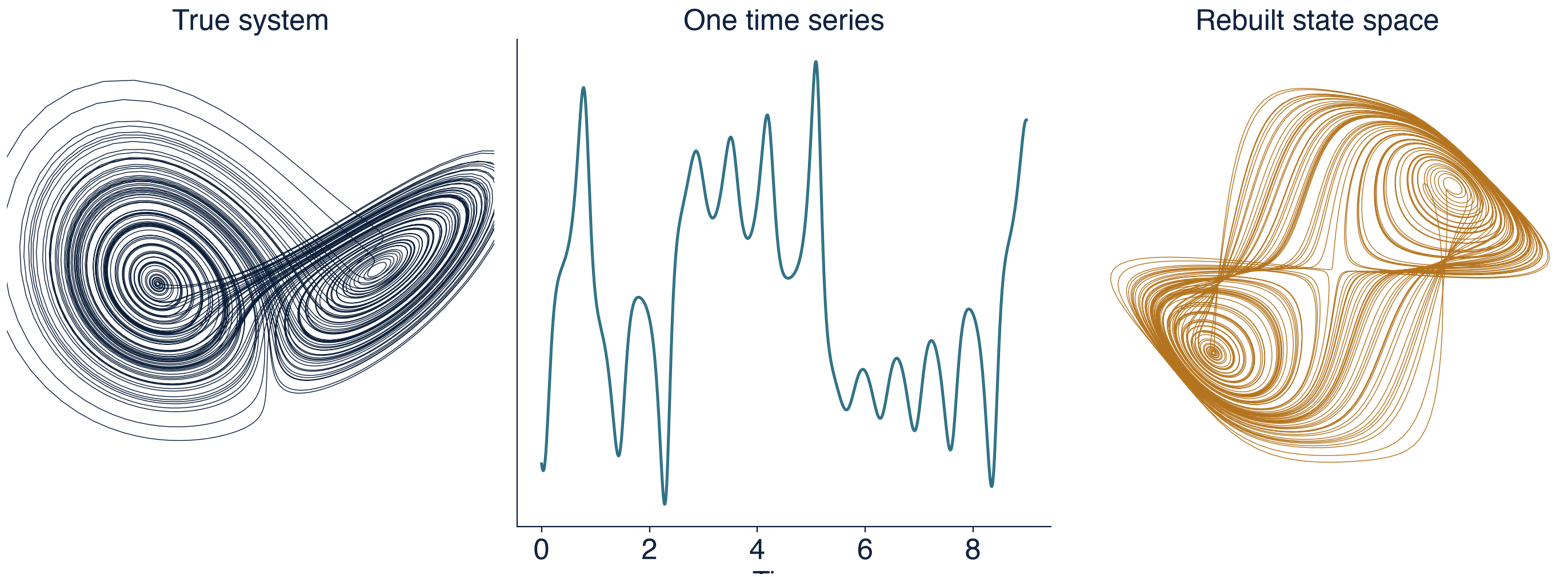
WSL

**SeeWandel-Climate** · Pomati lab and colleagues at Eawag
Luis J. Gilarranz · Stefanie Eyring · Stephan B. Munch · Tanya L. Rogers · Francesco Pomati
Data: IGKB, IBKF, University of Konstanz (D. Straile)

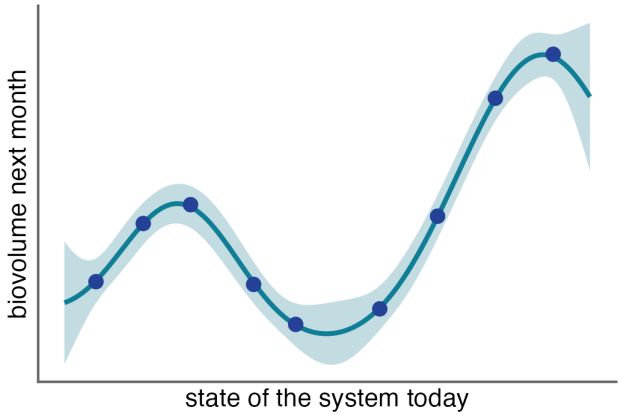
# Supplementary materials



# What is GP-EDM?



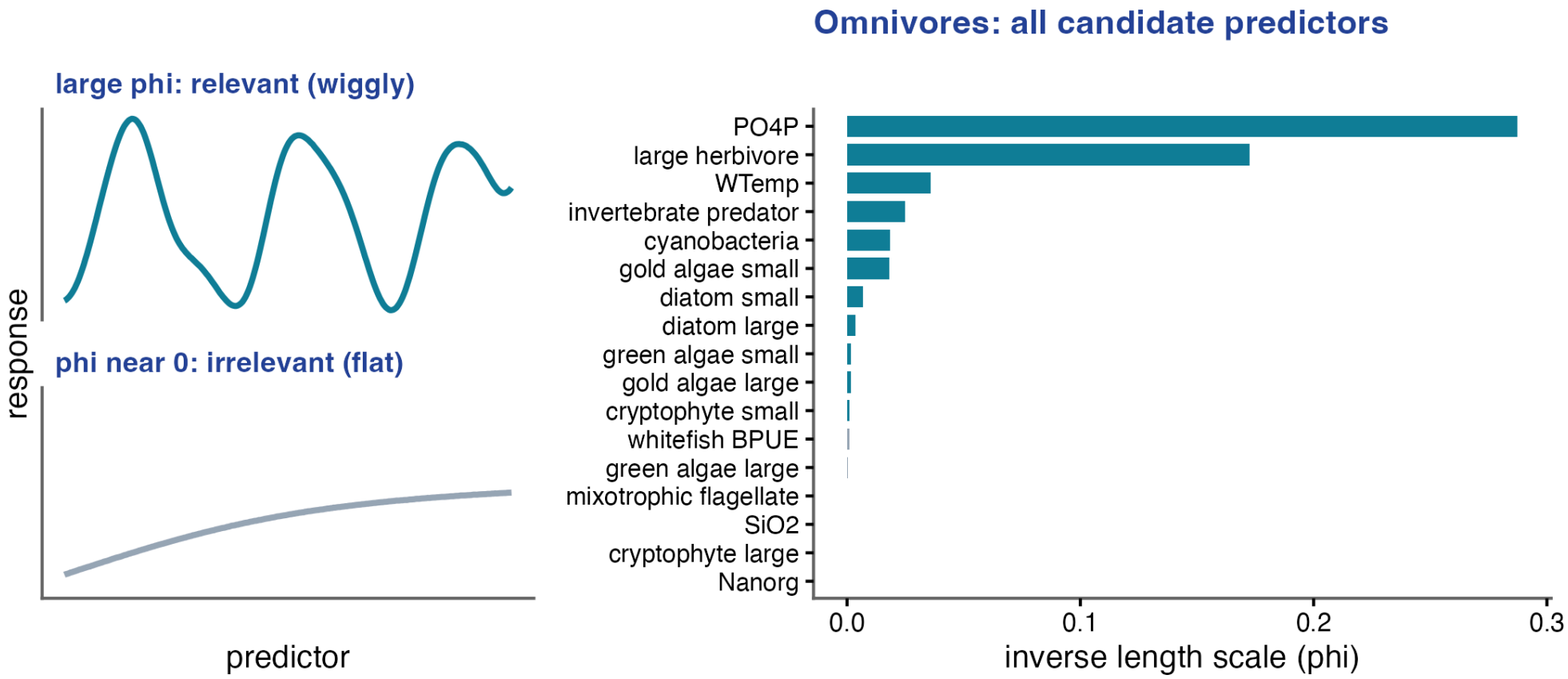

**Gaussian process: no fixed equation**




Next month = f(past states, drivers)

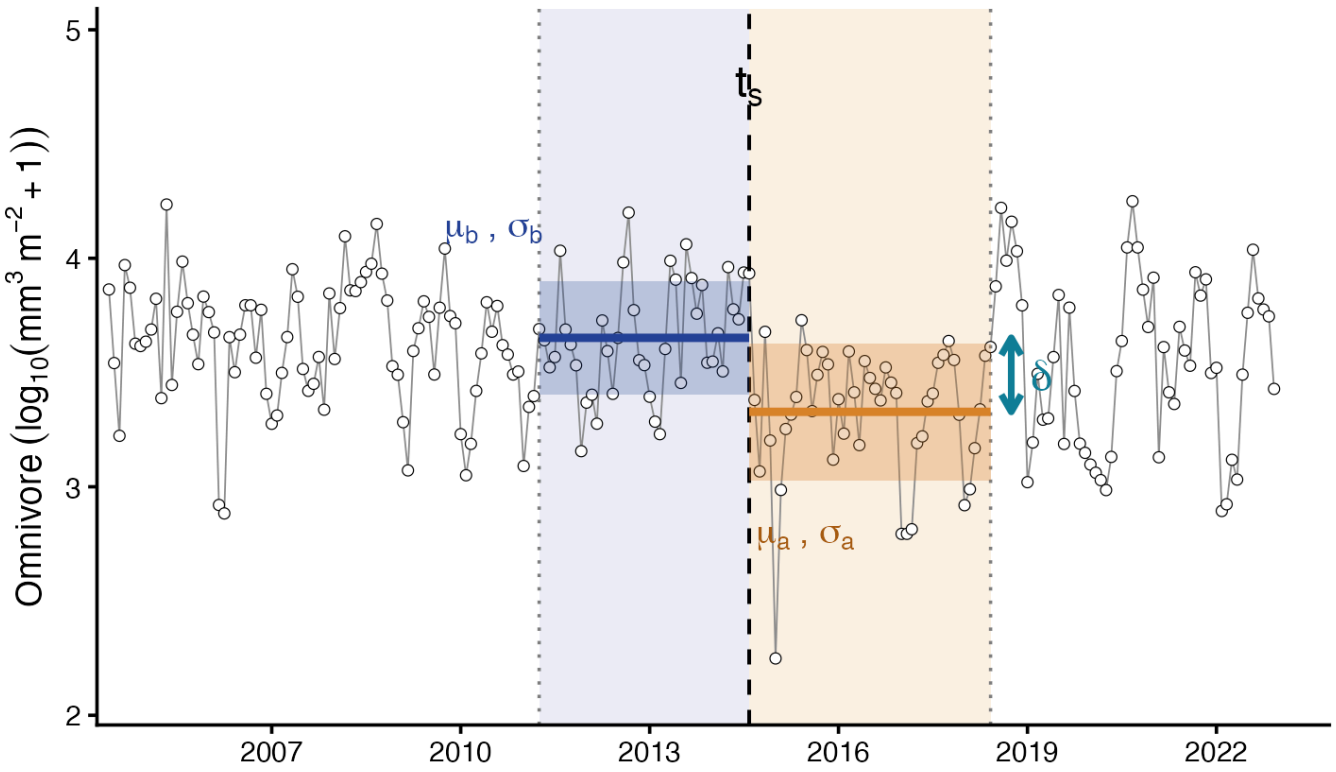
f learned from the data: Gaussian process

# Automatic relevance determination (ARD)




One length scale per predictor: **phi near 0** = predictor dropped

# Detecting regime shifts: BEAST and two criteria




**1** BEAST: candidate shift

$t_s$ : Pr(change point) = 0.87

**2** Difference of the two chunks

$\delta = \mu_a - \mu_b \quad 0 \notin CI_{95\%}(\delta)$

CI = [0.19, 0.42] ✓

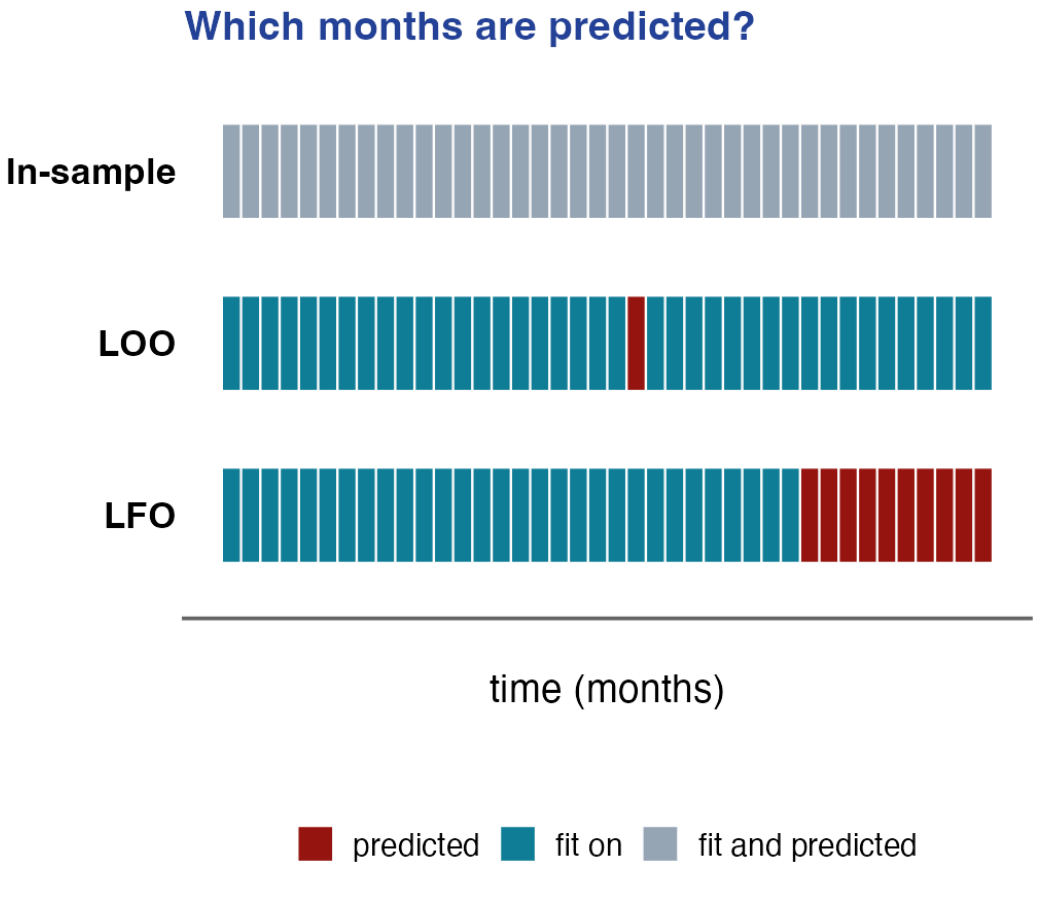
**3** Abruptness

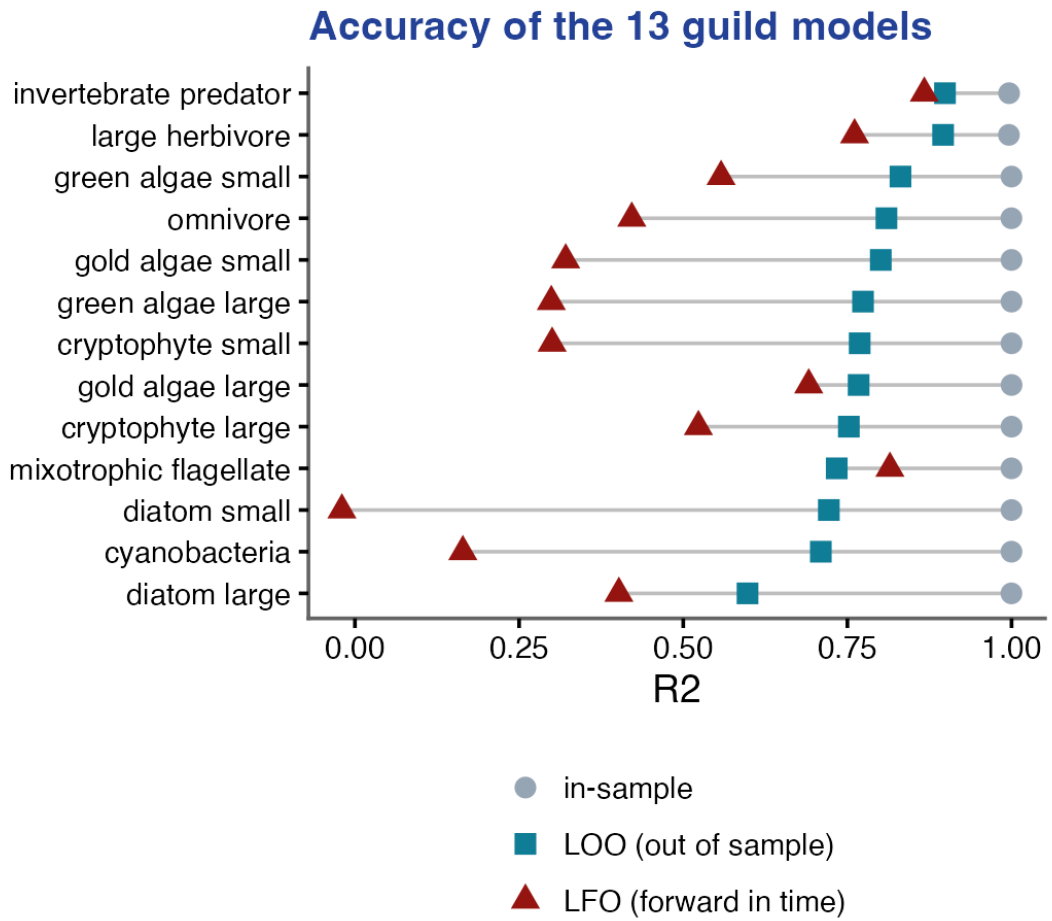
$$a = \frac{|\mu_a - \mu_b|}{\frac{1}{2}(\sigma_b + \sigma_a)} > 1$$

a = 1.19 ✓

Kept if 2 and 3 hold

# In-sample versus out-of-sample $R^2$




In-sample: $R^2 = 1$, a GP interpolates

LOO: one month left out

LFO: forecast the last 10 %

# Which genera? Zooplankton and mixotrophs



## Invertebrate predators

Pr

| | |
|---|---|
| *Leptodora* | 73% |
| *Bythotrephes* | 27% |



## Omnivores

Om

| | |
|---|---|
| *Cyclops* | 80% |
| *Mesocyclops* | 17% |
| *Acanthocyclops* | 1% |
| *Thermocyclops* | 1% |
| *Asplanchna* | 0% |

+ 4 more



## Large herbivores

He

| | |
|---|---|
| *Daphnia* | 91% |
| *Eudiaptomus* | 6% |
| *Bosmina* | 2% |
| *Diaphanosoma* | 1% |
| *Acanthodiaptomus* | 0% |

+ 6 more



## Mixotrophic flagellates

Mi

| | |
|---|---|
| *Ceratium* | 46% |
| *Gymnodinium* | 45% |
| *Peridinium* | 7% |
| other Flagellaten | 1% |
| other Phytoflagellaten | 0% |

+ 14 more

Top genera by share of biovolume, IGKB 1975–2022

# Which genera? Phytoplankton (1/2)



## Cyanobacteria

Cy colonial

| | |
|---|---|
| *Anabaena* | 44% |
| *Aphanizomenon* | 16% |
| *Planktothrix* | 9% |
| *Cyanothece* | 7% |
| *Aphanothece* | 6% |

+ 25 more



## Green algae, small

Gr1 ≤ 550 µm³

| | |
|---|---|
| Volvocales | 30% |
| unidentified algae | 24% |
| *Chlorella* | 21% |
| *Oocystis* | 7% |
| *Characium* | 3% |

+ 21 more

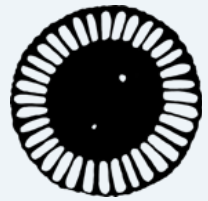


## Green algae, large

Gr2 > 550 µm³ or colonial

| | |
|---|---|
| other Chlorococcales | 23% |
| *Pandorina* | 18% |
| *Mougeotia* | 16% |
| *Chlamydomonas* | 10% |
| *Ulothrix* | 6% |

+ 46 more

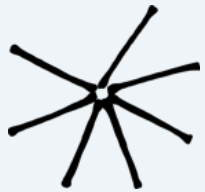


## Diatoms, small

Di1 ≤ 800 µm³

| | |
|---|---|
| *Stephanodiscus* | 81% |
| *Cyclotella* | 17% |
| *Achnanthes* | 1% |
| *Fragilaria* | 1% |
| *Nitzschia* | 0% |

+ 9 more



## Diatoms, large

Di2 > 800 µm³ or colonial

| | |
|---|---|
| *Fragilaria* | 50% |
| *Stephanodiscus* | 15% |
| *Asterionella* | 14% |
| *Cyclotella* | 10% |
| *Tabellaria* | 4% |

+ 18 more

Top genera by share of biovolume, IGKB 1975–2022

# Which genera? Phytoplankton (2/2)



## Gold algae, small

Go1 ≤ 250 µm³

| | |
|---|---|
| *Erkenia* | 63% |
| *Chrysochromulina* | 23% |
| *Chromulina* | 4% |
| *Ochromonas* | 2% |
| *Pseudokephyrion* | 2% |

+ 13 more



## Gold algae, large

Go2 > 250 µm³ or colonial

| | |
|---|---|
| *Dinobryon* | 42% |
| *Mallomonas* | 17% |
| *Uroglena* | 12% |
| Chrysophyceae | 12% |
| *Pseudopedinella* | 6% |

+ 9 more



## Cryptophytes, small

Cr1 ≤ 700 µm³

| | |
|---|---|
| *Rhodomonas* | 99% |
| *Plagioselmis* | 1% |
| *Cyathomonas* | 0% |
| *Cryptaulax* | 0% |
| *Chilomonas* | 0% |

+ 1 more



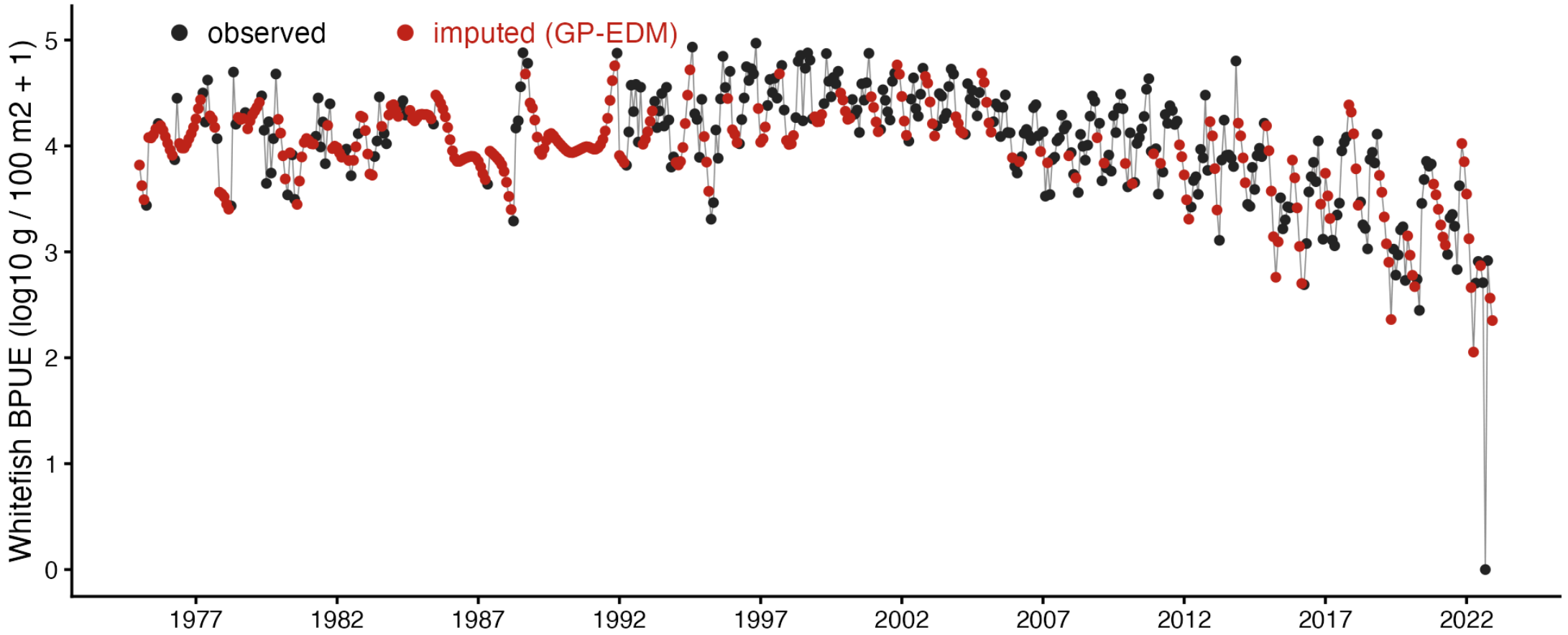
## Cryptophytes, large

Cr2 > 700 µm³

| | |
|---|---|
| *Cryptomonas* | 100% |
| other Cryptophyceen | 0% |

Top genera by share of biovolume, IGKB 1975–2022

# Whitefish time series: observed and imputed




287 of 576 months imputed (50 %)

GP-EDM, LOO $R^2$ = 0.55

# References


Eyring S, Merz E, Reyes M, Ntetsika P, et al. 2025. Distinct phytoplankton size classes respond differently to biotic and abiotic factors. ISME Communications 5: ycae148.

Gilarranz LJ, Narwani A, Odermatt D, Siber R, Dakos V. 2022. Regime shifts, trends, and variability of lake productivity at a global scale. PNAS 119: e2116413119.

Medeiros LP, Sorenson DK, Johnson BJ, Palkovacs E, Munch SB. 2025. Revealing unseen dynamical regimes of ecosystems from population time-series data. PNAS 122: e2416637122.

Merz E, Saberski E, Gilarranz LJ, Isles PDF, et al. 2023. Disruption of ecological networks in lakes by climate change and nutrient fluctuations. Nature Climate Change 13: 389–396.

Munch SB, Poynor V, Arriaza JL. 2017. Circumventing structural uncertainty: a Bayesian perspective on nonlinear forecasting for ecology. Ecological Complexity 32: 134–143.

Ntetsika P, Eyring S, Merz E, Reyes M, et al. 2026. Biotic interactions shape the realized niche of toxic cyanobacteria. Nature Ecology & Evolution 10: 1434–1448.

Takens F. 1981. Detecting strange attractors in turbulence. Lecture Notes in Mathematics 898: 366–381.

To S-W, Chakraborty S, Acevedo-Trejos E, Pomati F, et al. 2026. Future warming reshapes inorganic nutrient and plankton dynamics in a temperate lake. Limnology and Oceanography 71: e70447.

Lake Constance map (panel a): adapted from the published figure provided by the author. Panels b–d: IGKB monitoring data (phosphate, temperature) and IBKF whitefish catches, annual means 1975–2022.